

Brief — LAW 0AR — Law of Action–Reaction from Origin

A Bounded, Origin-Centered Reformulation of Action–Reaction for Classical and Non-Classical Physics

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Mathematical Note

This brief distinguishes between:

- **Structural Symmetry LAW 0AR — The Physical Law**

The structural principle describing origin-centered action–reaction symmetry.

- **SSMO-AR — The Mathematical Operator**

A purely symbolic, bounded mapping that computes a unitless structural pair:

$$\begin{aligned}A(F) &= Z(B(D(F))) \\ R(F) &= -A(F)\end{aligned}$$

Where **F** is a **nondimensional input proportional to the physical force** (the physical units are never passed directly into the mapping).

The operator does **not** redefine force, modify Newtonian mechanics, or replace the Third Law.

It simply provides a **bounded, origin-symmetric structural representation** that behaves consistently across classical and non-classical regimes.

1. Abstract

Sir Isaac Newton’s Third Law remains one of the most profound and elegant principles in physics.

In its classical form, Newton states:

**“To every action there is always opposed an equal reaction:
or the mutual actions of two bodies upon each other are always equal,
and directed to contrary parts.”**

This holds exactly in rigid, local, instantaneous interactions.

However, in many real systems — fluids, fields, plasmas, jets, granular media, distributed structures — the point-to-point interpretation becomes difficult to apply directly, even though momentum conservation remains exact.

LAW 0AR provides a modern, complementary interpretation of symmetry for such systems.

LAW 0AR — Core Statement

**Every physical interaction produces a dual, symmetric,
bounded pair of structural responses originating from
the same interaction source.**

This formulation:

- preserves Newtonian symmetry where classical assumptions hold,
 - extends interpretability to distributed and field-mediated interactions,
 - introduces no new units, forces, or constants,
 - provides a stable, bounded structural representation of interaction symmetry across classical and non-classical regimes.
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Note on the Designation of LAW 0AR as a Physical Law

At first glance, LAW 0AR may appear to be an analytical tool rather than a physical law.

This impression comes from its operational form

$$A(F) = Z(B(D(F))),$$

which resembles a computational pipeline.

Yet the underlying principle satisfies the criteria expected of a physical invariant:

universality, symmetry, boundedness, monotonicity, classical recoverability, and frame independence.

The operator is the mechanism.

The law is the structural principle:

“Every physical interaction produces a dual, symmetric, bounded pair of responses from a common origin.”

In classical rigid-body limits, LAW 0AR reduces exactly to Newton’s familiar behaviour:

$$\begin{aligned} \mathbf{A} &\approx \mathbf{F}_{AB} \\ \mathbf{R} &\approx -\mathbf{F}_{AB} \end{aligned}$$

preserving the Third Law where its assumptions hold.

In distributed or field-mediated systems — jets, pressure fields, granular flows, plasma interactions, momentum stored in EM fields — the classical point-force picture becomes difficult to apply locally, yet the origin-centered symmetry of LAW 0AR continues to hold.

Thus, LAW 0AR is presented not as a replacement for Newton’s formulation, but as a **structural symmetry law** that preserves the classical limit while offering a unified interpretive framework for complex, non-local interactions.

2. Why LAW 0AR Is Needed

Classical mechanics expresses action–reaction as a **pairwise, point-to-point exchange of forces**.

This works perfectly for rigid bodies and local interactions — the domain for which Newton formulated the law.

However, in many real systems **the interaction is not concentrated at a single point**, even though global momentum conservation remains exact.

2.1 Where the Classical Pairwise Picture Becomes Difficult

Examples include:

- **Rocket exhaust and jet propulsion** — momentum is distributed across billions of particles, not exchanged between two discrete bodies.
- **Aerodynamic lift** — forces arise from pressure fields over surfaces, not from direct contact between bodies.
- **Electromagnetic interactions** — momentum is stored and transferred through fields, not solely through mechanical contact.
- **Plasma and ion engines** — thrust emerges from continuous momentum flux rather than pairwise mechanical forces.
- **Fluid and turbulent flows** — force exchange occurs through distributed pressure, shear, and vorticity patterns.

Across these systems:

- Newton's Third Law holds **globally** (total momentum is conserved),
- but **the local equal-and-opposite pair is difficult to identify or express**.

LAW 0AR provides an **origin-centered symmetry representation** that remains fully consistent with classical mechanics while improving interpretability in distributed, non-local, or field-mediated interactions — **without altering physics in any way**.

3. What LAW 0AR Actually Is (and Is Not)

3.1 LAW 0AR is...

- a physical symmetry law,
- an **origin-centered formulation** of action–reaction,
- **classically consistent** — it reduces exactly to Newton's behaviour in rigid-body limits,
- a **structural symmetry principle** applicable even when forces are distributed or field-mediated,
- a **conceptual bridge** linking classical mechanics with complex multi-body, fluid, plasma, or field-based interactions.

LAW 0AR is the physical principle.

SSMO-AR is the mathematical operator that computes the unitless structural pair:

$$\begin{aligned}A(F) &= Z(B(D(F))) \\ R(F) &= -A(F)\end{aligned}$$

with **F** being **non-dimensionalized before entering the operator**.

3.2 LAW 0AR is not...

- not a computational transform,
 - not a structural mapping (that is SSMO-AR),
 - not a replacement for Newton's Third Law,
 - not a new physical force model,
 - not an engineering prediction formula,
 - not a modification of momentum conservation.
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4. The LAW Itself

LAW 0AR — Core Statement

“Every physical interaction produces a dual, symmetric, bounded pair of structural responses originating from the same interaction source.”

This is the physical principle:

- an **origin-centered symmetry law**,
- fully compatible with Newton’s classical formulation,
- **not** a new force law and **not** a modification of physical units.

LAW 0AR preserves the Third Law exactly where classical assumptions hold and provides an interpretive structure for systems where pairwise forces are difficult to localize.

Mathematical Structural Pair (SSMO-AR Operator)

The symbolic operator associated with LAW 0AR computes a **unitless, bounded structural pair**:

$$\begin{aligned}A(F) &= Z(B(D(F))) \\ R(F) &= -A(F)\end{aligned}$$

Where:

- **F is a nondimensional quantity proportional to the physical force** (physical units such as Newtons are *never* passed directly into the mapping),
- **A, R are unitless bounded outputs**,
- **D, B, Z** are dissipation, bounding, and collapse transforms.

These outputs:

- do **not** represent physical forces,
 - do **not** replace Newton’s formulation,
 - serve only as structural realizations of the symmetry expressed by LAW 0AR.
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Classical Correspondence

In the small-input (classical) regime:

$$\begin{aligned}A(F) &\approx F \\ R(F) &\approx -F\end{aligned}$$

This does **not** reproduce the magnitude of Newtonian forces.

It reproduces the **symmetry** and **directional pairing** that characterize Newton's Third Law in its rigid-body limit.

Thus:

- **Newton's Third Law** → **physical force symmetry**
- **LAW 0AR** → **structural symmetry expressed through a bounded, origin-centered pair**

Both remain consistent, with LAW 0AR providing a modern interpretive layer without altering the underlying physics.

5. How LAW 0AR Preserves Newtonian Symmetry

5.1 Classical Regime Consistency

When the transformation stages operate in their linear region:

$$\begin{aligned}D(F) &\approx F \\ B(x) &\approx x \\ Z(x) &\approx x\end{aligned}$$

Then:

$$\begin{aligned}A(F) &\approx F \\ R(F) &\approx -F \\ A + R &= 0\end{aligned}$$

Meaning:

- Newton's Third Law remains completely preserved at the level of symmetry
- LAW 0AR introduces **no modification** to classical mechanics
- In the rigid-body, small-input regime, the structural mapping behaves as a **normalized interpretive layer** over the same symmetry
- Only the symmetry is reproduced — not the physical magnitude of forces

This ensures that the classical Third Law remains the authoritative physical description wherever its assumptions hold.

5.2 The Complement, Not the Replacement

LAW 0AR becomes useful precisely when classical pairwise force assignment becomes difficult, such as:

- forces distributed over areas or volumes
- reactions not co-located with actions
- momentum transferred through fields
- fluid or plasma flow where reaction partners are continuous
- delayed or non-instantaneous interactions
- systems where the local action–reaction pair is ambiguous but global momentum conservation still holds

In these regimes, LAW 0AR:

- preserves Newtonian symmetry at the global level
 - provides a clean, bounded, origin-centered structural interpretation
 - does **not** alter the underlying physics
 - remains consistent with conservation laws and classical mechanics
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6. Mathematical Stability and Boundedness

LAW 0AR guarantees **stable, finite structural outputs** for every nondimensional physical input F :

$$A(F), R(F) \in (-1, +1)$$

(exact bounds depend on the chosen bounded transforms B and Z)

Key consequences:

- No divergence or blow-up at any magnitude
- No singularities
- Extreme physical forces collapse smoothly into a finite structural zone
- Perfect symmetry preserved at all scales:

$$A(F) + R(F) = 0$$

- Fully stable for numerical, computational, and real-time applications
- Physical forces themselves remain unbounded — only the structural layer is bounded

The SSMO-AR operator therefore provides a well-behaved, universal structural representation that remains stable regardless of the complexity or magnitude of the underlying physical interaction.

7. Real-World Validation Through Public Datasets

Independent public datasets from four unrelated physical domains were evaluated using the structural mapping:

$$\begin{aligned}A(F) &= Z(B(D(F))) \\ R(F) &= -A(F)\end{aligned}$$

where **F** is always a nondimensional input derived from the physical quantity, never the raw unit-bearing value.

Across all datasets, the mapping consistently exhibited:

- exact symmetry ($A + R = 0$)
- smooth, monotonic structural collapse
- bounded behaviour across all magnitudes
- preserved topology from small $\rightarrow$ moderate $\rightarrow$ extreme inputs

These evaluations do **not** reinterpret the physical force.

They simply demonstrate that the structural representation behaves coherently when real measurements from classical, distributed, and nonlinear systems are used as inputs.

7.1 Rocket Thrust Curves (Open-Aerospace Lambda-4S Dataset)

- The nondimensional thrust signal was passed through the D–B–Z sequence with no smoothing or filtering.
- Structural output maintained a constant bounded plateau during high-thrust phases, matching the empirical saturation of large-magnitude propulsion systems.
- Even during steep burnout decline, the structural output remained stable and well-behaved.

Outcome:

Shows that the OAR bounded-collapse behaviour parallels real thrust-curve saturation without modifying rocket physics.

7.2 Symmetric Aerodynamic Pressure Fields (NACA 0012 Public CFD Data)

- Distributed pressure values were nondimensionalised into effective interaction inputs F .
- The structural mapping preserved perfect symmetry across upper and lower surfaces.

- Topology matched the classical behaviour: strong leading-edge influence → smooth recovery.

Outcome:

Demonstrates that origin-based symmetry remains clean even when physical reactions are distributed through a continuous pressure field.

7.3 Asymmetric Aerodynamic Benchmarks (NACA 2412 Public CFD Data)

- Cambered-airfoil pressure distributions were processed into nondimensional interaction inputs.
- Structural outputs exhibited asymmetric lobes consistent with lift generation while still satisfying $A + R = 0$.
- High-gradient regions collapsed smoothly, while moderate regions preserved curvature.

Outcome:

Confirms interpretive stability in complex, non-uniform flow fields where classical pairwise force assignments are ambiguous.

7.4 Metal Stress–Strain Curves (Public CC0 Mechanical Dataset)

- Stress–strain data (elastic → yield → plastic → saturation) was converted to nondimensional F .
- Linear elasticity reproduced classical symmetry; plastic and saturation regions mapped into smooth bounded structural curvature.
- No discontinuities or instability occurred, even near yield peaks.

Outcome:

Shows that structural collapse naturally reflects the nonlinear, saturating nature of real material deformation.

Universal Observation Across All Datasets

$A(F) + R(F) = 0$ (verified at every point)

This exact symmetry — emerging from multiple unrelated empirical datasets — demonstrates that the OAR structural mapping is:

- mathematically stable
- physically interpretable
- domain-agnostic
- consistent across real-world conditions

LAW 0AR does **not** predict forces and does **not** alter physics.

It shows that when real measurements are passed through the mapping, the resulting structural behaviour remains coherent, bounded, symmetric, and topologically faithful — precisely the qualities expected of a valid structural extension of Newton’s symmetry principle.

8. What LAW 0AR Helps Clarify in Modern Systems

Newton’s Third Law remains exact within classical mechanics.

The challenges arise not in the law itself, but in how modern systems express interaction and momentum exchange.

LAW 0AR does **not** correct or modify physics.

It simply offers a clean structural lens for interpreting symmetry in environments where classical pairwise language becomes difficult to apply.

8.1 Distributed Interactions Need a Unified Structural Anchor

Context

In fluids, plasmas, pressure fields, and exhaust flows, reactions are spatially distributed rather than concentrated at a single point.

Clarification Offered by LAW 0AR

An origin-centered structural mapping expresses the same underlying symmetry in a form that remains meaningful even when the reaction is spread across a volume, field, or flow.

8.2 Numerical Models Benefit From Bounded, Stable Quantities

Context

Classical forces are unbounded — physically correct, but potentially destabilizing for numerical solvers, optimizers, or iterative models.

Clarification Offered by LAW 0AR

The SSMO-AR operator produces **bounded, monotonic, unitless structural outputs**, enabling stable symbolic comparisons while leaving all physical forces and units untouched.

8.3 Field-Mediated Interactions Need a Non-Local Interpretation Tool

Context

Electromagnetic, gravitational, and wave-mediated interactions conserve momentum, yet do not always present as a local equal-and-opposite force pair.

Clarification Offered by LAW 0AR

OAR’s symmetric structural outputs remain valid without requiring point contact, offering a clean structural representation for non-local momentum exchange while fully respecting classical conservation.

8.4 “Action at a Point” Is Not Always the Right Conceptual Lens

Context

Modern physics expresses interaction through momentum fluxes, stress tensors, and distributed pathways.

The classical phrase “*body A pushes body B*” is narrower than these realities.

Clarification Offered by LAW 0AR

Origin-centered symmetry mirrors how modern theory already treats interactions but does so in a **unit-free, bounded structural form** that complements existing formulations.

8.5 No Existing Structural Framework Spans All Domains

Context

Rigid-body mechanics, fluid dynamics, plasma physics, continuum mechanics, and field theory each use different mathematical constructs to express symmetry.

Clarification Offered by LAW 0AR

LAW 0AR acts as a single, domain-agnostic structural symmetry framework applicable across:

- rigid-body mechanics
- fluids and turbulence
- electromagnetic and gravitational fields
- plasma and ion interactions
- distributed or multi-node systems
- nonlinear and high-energy regimes

It introduces no new forces or constants — only a unified bounded structural representation.

Summary

LAW 0AR does **not** alter Newton's Third Law.

Instead, it clarifies symmetry in systems where classical pairwise language becomes difficult to apply, particularly when:

- reactions are distributed,
- forces propagate through fields,
- momentum flows through media, or
- numerical stability requires bounded symbolic quantities.

It is an interpretive framework — not a corrective one.

9. Scientific Position and Interpretive Boundary

LAW 0AR is defined with a clear scientific boundary:

- Newton's Third Law remains exact wherever classical mechanics applies
- no physical forces are altered
- no new units, constants, or predictions are introduced
- structural outputs (A, R) are dimensionless and independent of magnitude
- the SSMO-AR operator is a mathematical mapping — **not** a new physical force law

The classical regime remains fully intact.

The symmetry is exactly Newtonian whenever locality, rigidity, and instantaneous interaction hold.

Beyond classical conditions — where reactions become distributed, time-shifted, or field-mediated — LAW 0AR offers an origin-centered structural lens for expressing symmetry without modifying physics.

It is explicitly interpretive, not corrective.

10. Conclusion

LAW 0AR provides a modern, mathematically stable structure around one of physics' most enduring principles.

It:

- preserves Newtonian symmetry exactly
- clarifies interaction symmetry when pairwise forces are hard to isolate
- remains fully compatible with all established physics
- yields bounded, stable outputs ideal for modeling and computation
- demonstrates coherent behaviour across real public datasets
- avoids all claims of prediction or physical alteration

LAW 0AR is not a new physical theory.

It is a structural complement that clarifies symmetry while respecting classical mechanics entirely.

It stands as a contemporary companion to Newton's insight — built with:

- clarity,
 - bounded structure,
 - and disciplined mathematical framing.
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